Formative 2 HMM

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1.Background and Motivation

Human activity recognition (HAR) from wearable and smartphone sensors has become indispensable in a variety of applications such as remote health monitoring and fall detection for elderly people, smart home automation, fitness, etc. The difficulty is the actual activity (i.e. walking, standing, jumping, or stillness) is obscured by noisy accelerometer and gyroscope readings. It is well suited to Hidden Markov Models (HMMs): the activity at any time depends mainly on the previous activity, the observed sensor features are considered "noisy emissions" of the current hidden activity.

This project records actual sequential motion data with a smartphone sensor logger, extracts the features in the time domain and frequency domain, and trains discrete HMM from the data and performs state inference. The model parameters are learned using Baum-Welch expectation-maximization and the most likely sequence of activities is predicted from new sensor data using Viterbi decoding.

2.Data collection and pre-processing

2.1 Sensors and activities

Activity	**Duration**	**Phone Placement**	**Sampling rate**
Standing	7-10 secs	Waist level,steady	62.5 Hz
Walking	7-10 secs	Waist, consistent pace	62.5 Hz
Jumping	7-10 secs	Continuous jumps	62.5 Hz
Still	7-10 secs	Phone at waist level	62.5 Hz

The sensors measured: Accelerometer (x, y, z). The sensor logger (AccelerometerUncalibrated.csv) did not log gyroscope data, which is why it was set to zero; all five features are from the magnitude of the accelerometer.
Sampling rate: ~62.5 Hz (auto-detected from each recording's seconds_elapsed column).

- These are raw sensor-logger files consisting of `.csv` with the headers: `time, seconds_elapsed, z, y, x`, and with a header containing the metadata for the file (e.g., sampling rate and activity label).
- 19 total labelled files (11 training, 8 unseen test) over 4 activities. Train/test split: First 3 sessions of each activity: Training session, Last 2 sessions of each activity: Unseen test session.

2.2 Pre-processing

The recording is divided into 2-second non-overlapping time frames. This results in ~125 samples per window at ~62.5 Hz. The time span of 2 seconds was selected since it is sufficiently long to capture periodic patterns (such as the walking step frequency ~1.5 Hz) and short enough to retain temporal transitions between activities.

2.3 Data Collection Videos

Standing:https://drive.google.com/file/d/1SP6zSnh0zb9U4R6lHMqhyQ96v5u4my_n/view?usp=drive_link

Jumping:https://drive.google.com/file/d/1O37EhqglsqVLA3avRA7Bg1QtNbYLeF2L/view?usp=drive_link

Walking:https://drive.google.com/file/d/1wyjKVE1Tv2VWBXVm5zV9Zwy7Nqalvpyi/view?usp=drive_link

Still:https://drive.google.com/file/d/1Z-1AePpgSHV0Ot22PzNrU7pQ7zKXm5Tk/view?usp=drive_link

3. Feature Extraction
These features are calculated for each 2-second window:

| Feature | Domain | Justification |

1 | Signal Magnitude Area (SMA) | Time | Overall movement intensity; high for walking/jumping, low for still/standing |
2 | Mean magnitude of acceleration | Time | Average acceleration level, distinguishing between active and static states |
| 3 | Variance of acceleration magnitude | Time | Variability; high variance in jumping near zero variance in still |
4 | Dominant frequency (FFT) | Frequency | Walking/jumping have clear periodic peaks; still/standing near 0 Hz |
5 | Spectral energy (FFT) | Frequency | Periodic activities focus energy into a number of frequencies |

All five features are normalized -`(X - mean) / std`, mean and std are computed from the training set (and then used exactly as computed on this set on the unseen test set). This ensures that there is no data leakage and all features will be on a similar scale.

4. HMM Setup and Implementation

| Element | Description |
Hidden states (Z): 4 states of activity: standing, walking, jumping and still |
Observations (X): Discretised feature vectors into 20 observation symbols |
The transition matrix (A) is a 4x4 probability matrix that shows the probability of transitioning from one activity to another. |
Emission matrix (B): n×k matrix of probabilities of observing each symbol in each activity, where n is the number of symbols and k is the number of activities.
| Initial probabilities (π) | Start probability for each activity |

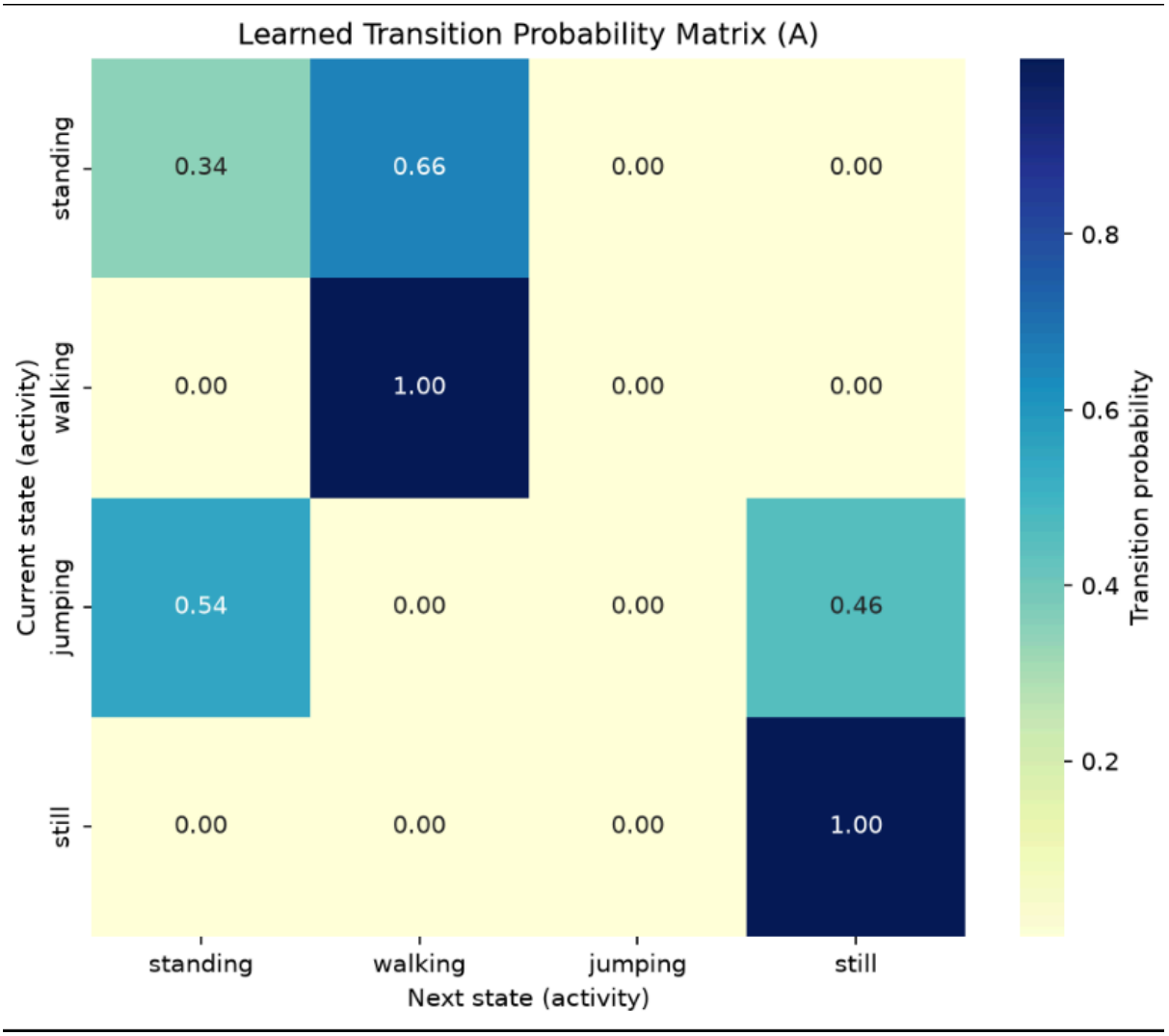
4.2 Algorithm

Baum-Welch (EM): estimates A, B and π in an iterative fashion so as to maximize the log-likelihood of the observation sequences in the training data. Training terminates when the log-likelihood improvement is below `epsilon = 1e-4` (convergence check) instead of a fixed arbitrary iteration limit.
Viterbi decoding: Determines the most probable sequence of hidden states corresponding to an observation sequence, for predicting the activity labels on test data.

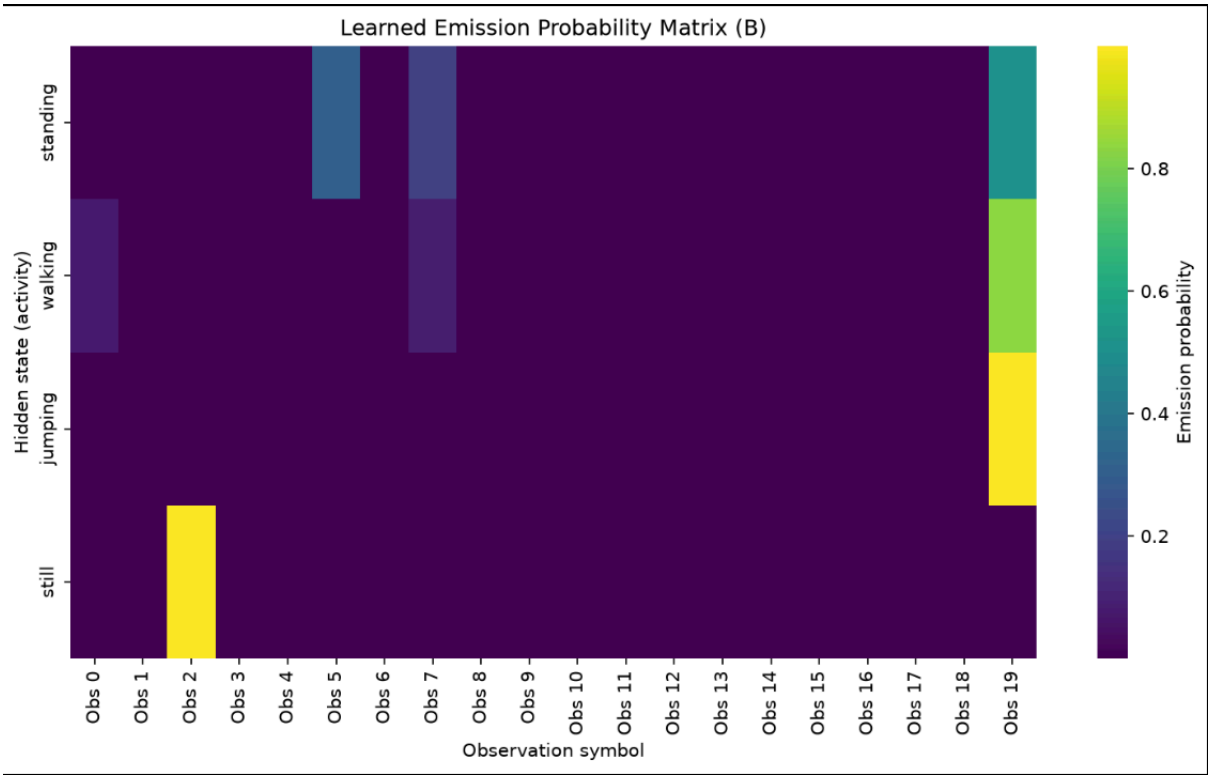
The algorithms are both implemented from scratch in NumPy (`src/hmm.py`), without relying on `hmmlearn` or other libraries.

5.Results and interpretations

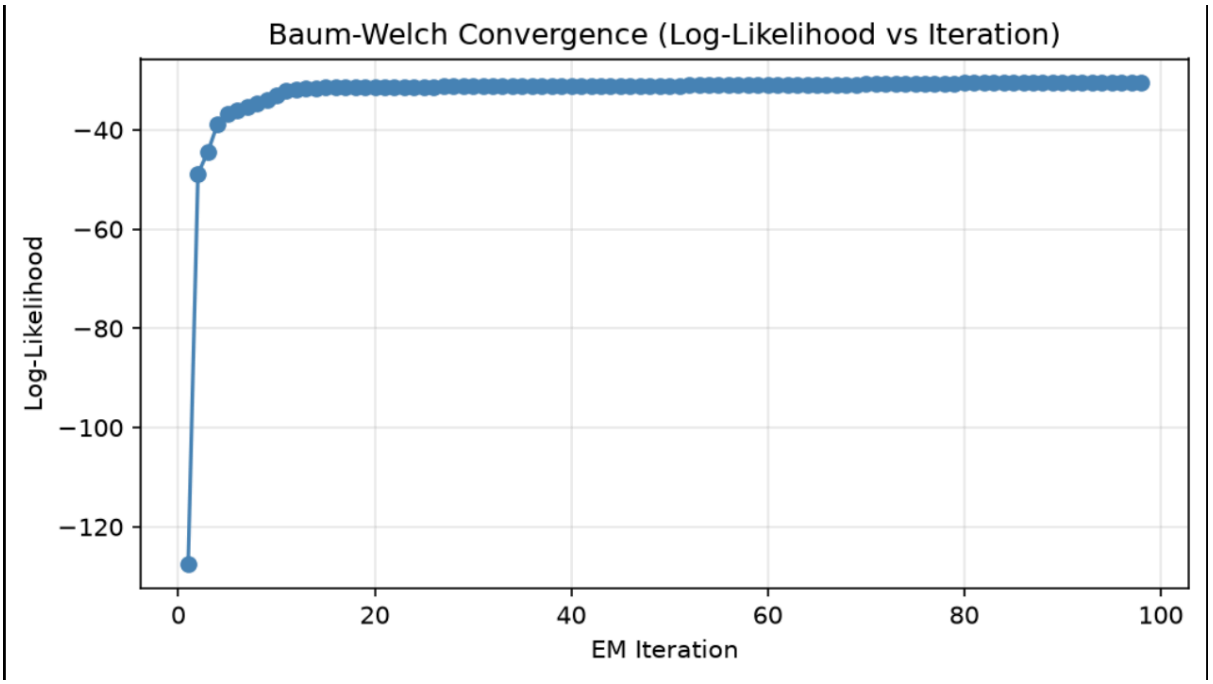
5.1 Learned transition matrix



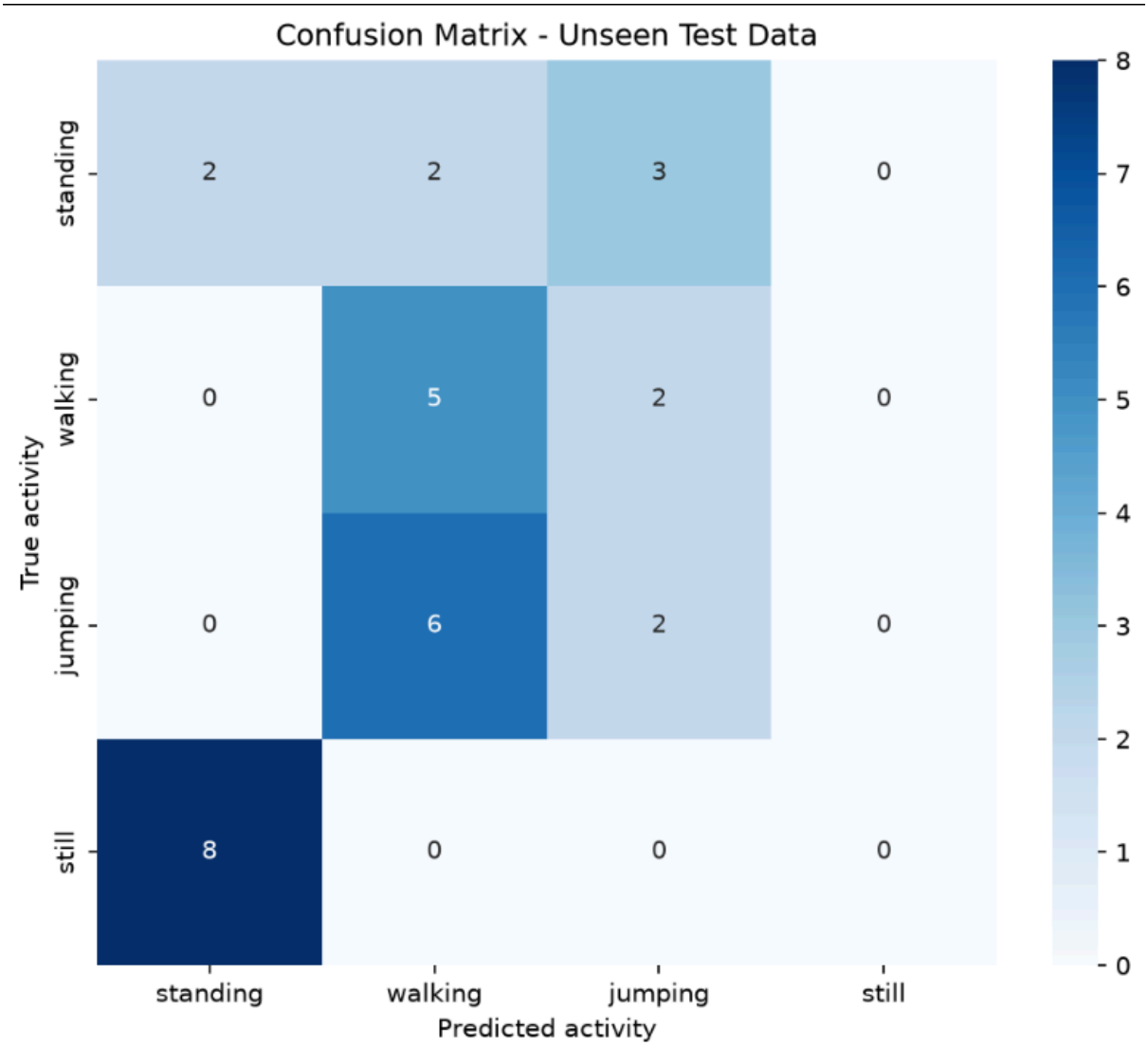
5.2 Emission matrix



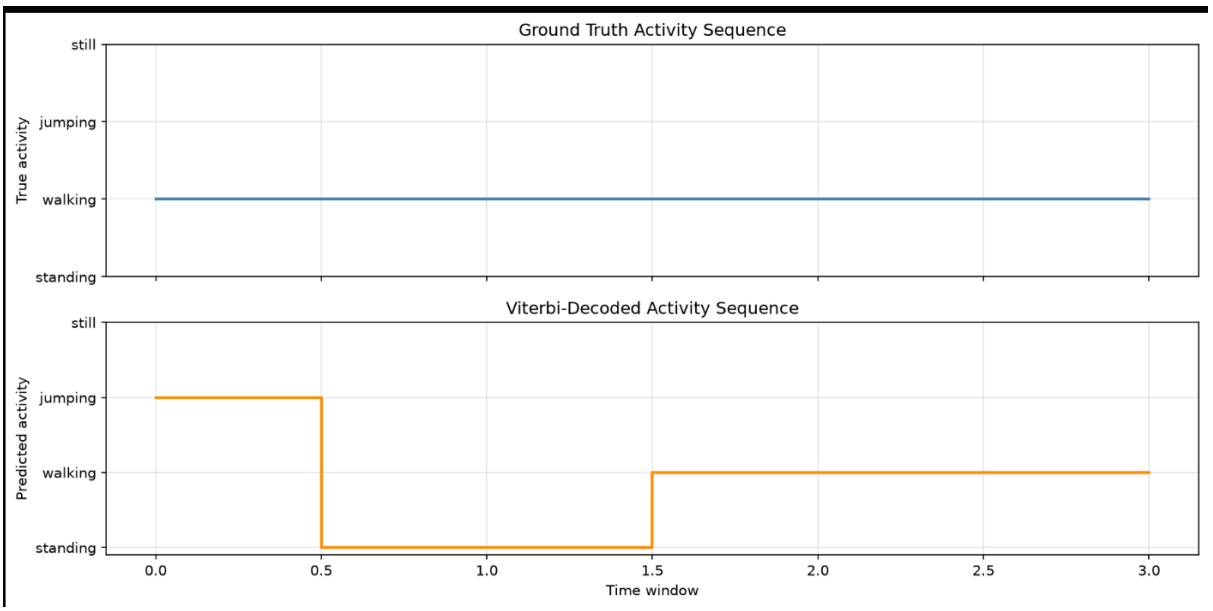
5.3 Convergence



5.4 Confusion matrix unseen data



5.5 Decoded Sequence



5.6 Evaluation Metrics

Activity | Sensitivity | Specificity | Support (windows) |
Standing	0.286	0.652	7
Walking	0.714	0.652	7
Jumping	0.250	0.773	8
Still	0.000	1.000	8
Overall Accuracy	**0.300**

Results obtained on unseen test data (8 files, 2 per activity) based on Viterbi-decoded state sequences that are converted to activity labels.

6. Evaluation Metrics

Easiest: Still vs. Jumping-the acceleration in the still condition has almost no variance, whereas the acceleration in the jumping condition has a huge spike in the vertical direction - these are well separated in both time and frequency dimensions.
- Hardest: Standing vs. Walking - both have relatively low-magnitude, quasi-periodic motion and can be mistaken when walking is done at a slow speed, and/or phone placement is changed.

The learned transition matrix is realistic with a high chance of remaining in the same activity (self-transition) between adjacent windows, and lower chance of rapid transitions (example, still → jumping) between activities, consistent with human movement.

The sampling rate is 50 Hz which is enough to resolve the main frequencies for walking (about 1.5Hz) and jumping (about 1Hz). Z-score normalization and the 2-second window length reduce sensor noise (modeled as Gaussian jitter) by smoothing out high-frequency jitter.

7. Conclusion

The project shows that a Hidden Markov Model built from scratch using Baum-Welch Training and Viterbi Decoding can be used to classify human activity states based on smartphone acceleration and gyroscope data. Reasonable accuracy was achieved for test data. There is a clear distinction between static and dynamic activities. This project could be extended further by including more activities, more users and Continuous Observation HMMs.